

Dates Lab #	LECTURE Monday, 1-3pm IRC G1	LAB MODULE ACTIVITIES T/W/Th/F 1-5pm Biological Sciences 3145	NOTES & FORMAL ASSESSMENTS
Jan 5-9	Course Orientation Research project overview	No in-person labs	
LAB 1 Jan 12-16	Lab 1 overview (Aseptic work/bacterial work & Plasmids) Reading scientific papers activity Part 1	Bacterial culture aseptic technique - Bacterial transformation - Plasmid preparation	Self-Assessment 1 Due: Jan 12 th Lab Safety Quiz Lab 1 Quiz Due: Jan 12 th
LAB 2 Jan 19-23	Lab 2 overview (Mammalian cell culture) Reading scientific papers activity 2	Mammalian cell culture and aseptic technique - Basic tissue culture skills - Subculturing adherent cells; counting cells - Phase microscopy Record results from lab 1	(Last day to drop without a 'W') Lab 2 Quiz Due: Jan 19 th
LAB 3 Jan 26-30	Lab 3 overview (Cell stimulation SDS PAGE) Reading scientific papers activity 3	Cell stimulation, Protein Extraction and quantification; Cast resolving gel for SDS-PAGE	Lab 3 Quiz Due: Jan 26 th
LAB 4 Feb 2-6	Lab 4 Overview (SDS-PAGE, Staining)	SDS-PAGE: Cast stacking gel, prep protein, run gel, transfer Fluorescent labeling cultured cells Formative feedback – BSC Aseptic technique	Lab 4 Quiz Due: Feb 2 nd
LAB 5 Feb 9-13	Lab 5 overview (Western blotting, Fluorescence microscopy)	Western blotting Fluorescence microscopy Formative feedback – BSC Aseptic technique	Lab 5 Quiz Due: Feb 9 th Self-Assessment 2 Due: Feb 15 th Annotated Bibliography Worksheet and Concept Map Due: Feb 15 th
Feb 16-20	FAMILY DAY/READING WEEK		
LAB 6 Feb 23-27	Making figures and writing figure legends Data analysis workshops: Image J – Analyzing images Image J – Western blot quantification	Practical Assessment – Tissue culture work in the BSC	IN LAB Formal assessment – BSC Aseptic technique
LAB 7 Mar 2-6	Lab 7 overview (ELISA)	ELISA	Lab 7 Quiz Due: Mar 6 th
LAB 8 Mar 9-13	Lab 8 Overview (RNA Extraction and qPCR) Data analysis Workshop - ELISA	RNA Extraction and qPCR	
LAB 9 Mar 16-20	Lab 9 overview (Splenic cell isolation and surface antigen labeling) Lab 10 overview (Flow cytometry) Data analysis Workshop: qPCR Data analysis and formal figure support	Splenic cell isolation and surface antigen labeling	Formal Figure and figure legend (individual) Due: Mar 22 nd
LAB 10 Mar 23-27	Scientific Writing Storyboarding our manuscript - Recap of lab activities	Field trip to the LSI Flow Cytometry Facility Flow cytometry and Data Analysis workshop – (FlowJo)	
NO LAB Mar 30-Apr 3	Flow Cytometry debrief Flow Cytometry Assignment support	FLEX WEEK – NO LABS In lab – data analysis and writing support	
NO LAB Apr 6-10	Preparing your manuscript Peer Review Session	FLEX WEEK – NO LABS In lab – data analysis and writing support	Peer Assessment (Pair work survey) Due: April 10 th Flow Cytometry Assignment Due: April 12 th Manuscript (in pairs) Due: April 19 th (date may change depending on final exam date)
Final Exam period Apr 14-25	Final exam date TBA (exam schedule released Mid-Feb)		Due on final exam day: Lab Notebook Self-Assessment 3